

CHAPTER 2

Why Don't Media Violence Effects Look the Same on Everyone?: Developmental Approaches to Understanding Media Effects

Douglas A. Gentile

When the first edition of this book came out, several violent events had recently occurred that seemed to implicate media violence as a cause. These included a 12-year-old boy brutally killing a 6-year-old girl by imitating professional wrestling moves (“Stiff Sentence,” 2001), the Columbine High School massacre, carried out by two adolescents who had practiced their plan by playing violent video games (Gibbs & Roche, 1999), and a 17-year-old setting himself on fire while being videotaped in an attempt to duplicate a stunt called the “human barbecue” from the MTV show *Jackass* (Brachear, 2002). At that time, we argued that these highly publicized stories demonstrated how *not* to think about media effects because they oversimplify complex situations (Gentile & Sesma, 2003). In the past decade, however, the public’s understanding of media violence seems to have actually gotten worse. Incidents linking media violence to extreme criminal acts, such as the mass murder at an Aurora, Colorado, movie theater in 2012 (www.cnn.com/SPECIALS/us/colorado-shooting), continue to be reported as if media violence was responsible. These claims understandably beget an equal, but opposite, reaction, with people claiming that they played violent games and haven’t shot anyone so therefore they must have no harmful effect, or might even have a helpful effect (e.g., White, 2013). Given the increasing polarization of American society (Dandekar, Goel, & Lee, 2013), perhaps this controversy is to be expected. Nonetheless, even the scientific dialogue appears polarized, with some reviews claiming that the debate should be “over” (Anderson et al., 2003), and others claiming that the researchers are trying to incite a “moral panic” (Ferguson, 2010).

To be able to find the middle ground and to understand media effects on children more fully, it is necessary to dispel some myths. There are many common beliefs about media effects that despite being simple, persuasive, and sometimes partly true, actually hamper comprehension of how media influence youth.

NINE MYTHS ABOUT MEDIA EFFECTS

Myth 1. Media Effects Are Simple and Direct

Examples that appear in the news seem to demonstrate simple and direct effects of children imitating what they have seen. Yet the effects of media are not simple; neither are they usually direct. Most media effects are cumulative and subtle, even when they are specifically designed to influence behavior. This subtlety masks remarkable power and persuasiveness. For example, research on advertisements demonstrates that attitudes and purchasing behavior can be altered by as few as two or three exposures to an advertisement (Woodward & Denton, 2000). Yet, as we watch or drive past

advertisements, we don't *feel* our opinions changing. The effects of media usually happen at a level at which we are not consciously aware. We rarely notice that advertisements affect us, partly because ads are generally presented as entertainment, so that viewers are less likely to notice any effect, or to resist their messages. Research suggests that the more one participates in this charade, claiming that advertisements don't affect "me," the *more* likely one is to actually be affected (Greene, 1999).

Thus, to the extent that we expect media effects to be simple and direct, we are probably failing to notice the strongest and most powerful media influences.

Myth 2. The Effects of Media Violence Are Severe

Each of the events mentioned at the beginning of this chapter is extreme, and yet most people who watch media violence never seriously injure other people or themselves. Since media violence does not make them commit the same kind of violence, many people draw the inference that it has no effect on them or on most other people. Potter (1999) noted that there is a flaw in this thinking: "People know that others are committing violent acts, but they also know that they personally have never committed any atrocities. The problem with this reasoning is that people equate [media] effects with atrocities" (p. 122). Watching violent media can have many effects, and we should not expect that exposure to media violence will cause people to begin killing each other. The vast majority of research on media violence demonstrates that it influences aggressive thoughts and behaviors, which in turn increase the odds of an aggressive response when provoked. Nonetheless, this aggressive response is most likely to be low-level aggression, such as verbal or relational aggression (see Coyne & Stockdale, this volume). In fact, the largest effect of media violence is probably not illustrated by individual examples of violent behavior, but by the "culture of disrespect" it has fostered and nurtured (Walsh, 2001). Interpersonal violence is just the endpoint on a continuum of disrespectful behavior. As [Figure 2.1](#) shows, for every one example of a school killing, there were over 7,000 serious injuries, 28,000 thefts, 44,000 physical fights, and 500,000 reports of bullying (B. Modzeleski, personal communication, January 9, 2003). Killing someone is just the most visible part of the phenomenon—there is a great deal of aggressive behavior that is not so extreme.

Figure 2.1
Incidents of Violent Behaviors in American Schools, 1998



This highlights an important point of agreement within what appears to be disagreement. The critics of media violence research almost always focus on "violence" rather than the less extreme and more

inclusive term “aggression.” Most of the researchers agree with the critics that there is very little evidence that media violence will cause real world *violence* (defined as physical aggression that would cause severe bodily harm or death). They also agree, however, that there is a lot of evidence that media violence can cause real world *aggression*. This point is often missed in the public dialogue.

Although most of the public concern is about behaviors, violent media affect us much more broadly. Any time that you have laughed, felt excited, become scared or otherwise aroused while watching a violent movie, *you have just been affected*. Positive and negative emotional and physiological reactions to violent media are media effects. Clearly, many people seek out this type of stimulation. After all, who wants to watch a “boring” movie? Violent media have many effects, including emotional, physiological, cognitive, attitudinal, and behavioral effects (see [chapters 3, 4, 5, 6, 8, and 11](#) for more details). To the extent that we expect media effects to be exhibited through atrocities, we are ignoring where the effect really happens.

Myth 3. Media Effects Are Obvious

In each of the examples at the beginning of this chapter it was obvious (often from the perpetrator’s own admission) what media product had influenced the subsequent behavior. This leads many to expect that such links should usually be obvious and to take the absence of a direct and obvious link as evidence that no media effects should be implicated. Because the effects of violent media are usually indirect, subtle, and cumulative (and thus not obvious), many people then argue that researchers and policymakers are trying to find an easy “scapegoat” to explain violent behaviors. Indeed, even when the link is obvious, many people make this argument; the following anonymous quote was posted in response to the *Jackass* copycat burning: “TV shows are not responsible for copycat attempts of dangerous stunts they portray. ... Blaming TV shows for the actions of minors is just passing the buck” (“Texas talkback,” n.d.).

Humans are not good at detecting small changes across time, which is how the media tend to have their effects. Furthermore, media violence influences psychological variables, such as attitudes, beliefs, feelings, scripts, expectations, biases, and perceptions, none of which are easily visible. Nevertheless, cause-effect relationships need not be obvious to be significant. Most people accept that smoking causes lung cancer, even though the effect is subtle and cumulative (for a description of the many parallels between smoking and media violence, see Bushman & Anderson, 2001). One cigarette does not change a person’s health in any particularly noticeable way, but years of smoking can have dire consequences (although not for all people). To the extent that we expect media effects to be exhibited in an obvious manner, we are missing opportunities to see other, less obvious and perhaps more pervasive effects.

Myth 4. Violent Media Affect Everyone in the Same Way

Many people assume that media violence effects must be unidimensional—that is, that everyone must be affected in the same way (by becoming more aggressive). Although that is one of the documented effects, it is not the only one. Meta-analyses (studies that analyze data presented across large numbers of studies) have shown that there are multiple effects. Some of these are at the level of emotions, some at the level of thoughts, some are about attitudes, some are about behavior, and so forth (see Potter, 1999, and this volume, for a listing of about 17 different scientifically documented effects). These multiple effects have been grouped into four, called the *aggressor effect*, the *victim effect*, the *bystander effect*, and the *appetite effect* (Donnerstein, Slaby, & Eron, 1994).

The *aggressor effect* describes how children and adults who watch a lot of violent entertainment tend to become more willing to behave aggressively when provoked, becoming meaner and more aggressive.

The *victim effect* describes how children and adults who watch a lot of violent entertainment tend to see the world as a scarier place, become more scared, and initiate more self-protective behaviors

(including going so far as to carry guns to school, which, ironically, *increases* one's odds of being shot).

The *bystander effect* describes how children and adults who watch a lot of violent entertainment tend to habituate to gradually increasing amounts of violence, thereby becoming desensitized, more callous, and less sympathetic to victims of violence (both in the media and in real life).

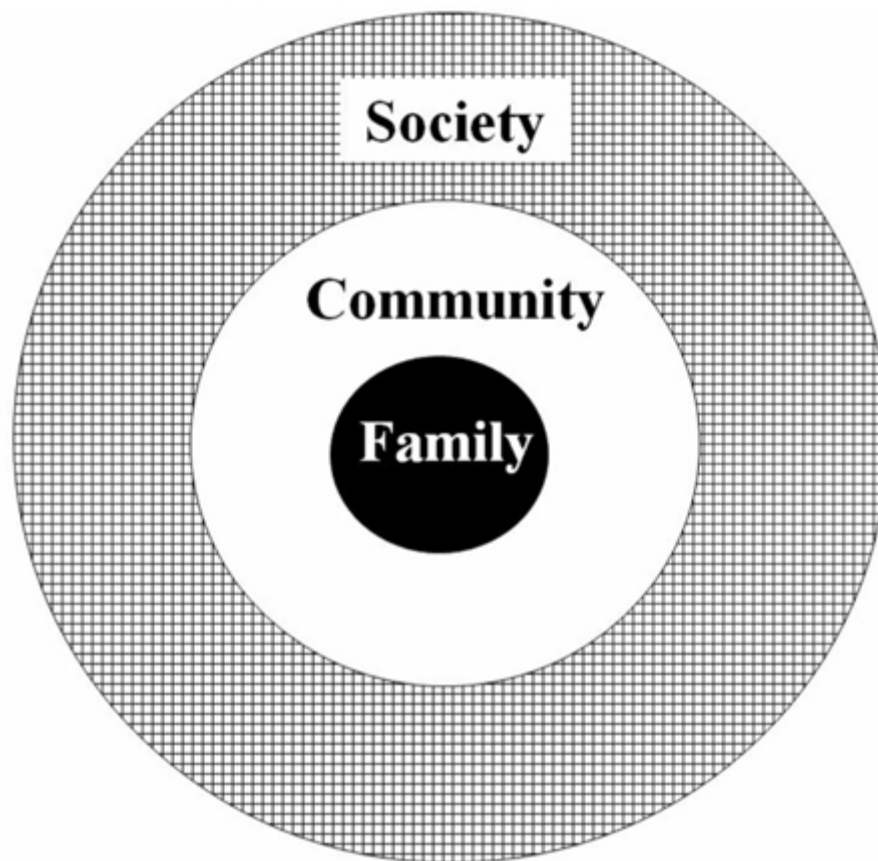
The *appetite effect* describes how children and adults who watch a lot of violent entertainment tend to want to see more violent entertainment. Simply put, the more one watches, the more one wants to watch.

These effects are well documented in hundreds of studies. What is less well known is which people are more prone to which effects (although these effects are not mutually exclusive). In general, females tend to be more affected by the *victim effect*, whereas males tend to be more affected by the *aggressor*, *bystander*, and *appetite effects*. But it is still unclear how to predict exactly how any given individual will be affected by any given media violence presentation. However, the fact that we cannot yet make this prediction very well should not be taken as evidence that there is no effect. Notice that just because everyone is not affected in the same way does not mean that every viewer is not affected.

To understand where children learn their attitudes, values, and patterns of behavior, we can consider the effects of various proximal and distal sources of influence (see [Figure 2.2](#)). The family has the most direct access to the children, and children clearly have their attitudes, values, and behavior patterns shaped and modified by their families. The behaviors defined as “normal” within each family affect the behaviors of the individuals within that family. Beyond the level of the family, the norms of the community affect the norms of the families and individuals within it. Beyond the level of the community, the norms of society affect the norms of the communities, families, and individuals within it. The media operate at this societal level, and media effects can be seen at all levels. Thus, the media can affect us not only one on one when we are watching TV, for example, but they also affect us by affecting the norms, expectations, and patterns of behavior of our families and communities. This is another aspect of the media's subtlety—they can affect us from multiple directions at once. Although this makes it likely that everyone will be affected by violent media in some way, it also makes it likely that the effects will not be identical for all people.

Figure 2.2

Multiple Spheres of Influence on Children



Myth 5. Causality Means “Necessary and Sufficient”

Determining if and when something “causes” something else is a problem that has plagued philosophers and scientists for centuries. In the social sciences, it is a surprisingly complex problem to solve. For many people, however, it has become oversimplified—something is a cause if it can be shown to be necessary and/or sufficient as a precursor. This position has been used to argue against the effects of media violence. Ferguson (2002), in a response to Bushman and Anderson’s (2001) meta-analyses of media violence and aggression, stated that:

- (a) because humans have always been violent, “violent media, then, are not a necessary precursor to violent behavior” (p. 446), and
- (b) because many people who are exposed to media violence never commit violent behavior, “violent media, then, are not sufficient to cause violent behavior” (p. 446).

This argument seems, on its surface, to be reasonable. Yet this argument actually betrays a grossly oversimplified idea of causation. Consider, for example, a rock on the side of a hill. Assume that you give the rock a push and it begins rolling down the hill. Did you cause the rock to roll down the hill? Based on the argument laid out above, you did not. Rocks have rolled down hills for eons without someone coming along and pushing them. Therefore pushing it is not necessary. Furthermore, many rocks that are pushed do not roll down hills. Therefore, pushing the rock is not sufficient. Although pushing the rock was neither necessary nor sufficient to make it roll down the hill, that does not mean that pushing was not a cause of the rock’s beginning to roll.

Most complex issues of interest (such as aggressive behavior) are multi-causal. In the present example, many other issues interact to determine whether the push you gave to the rock causes it to roll down the hill: the force of gravity, the mass of the rock, the shape of the rock, the friction of the hill surface, the slope of the hill surface, the direction of the push, the force of the push, how deeply the rock is embedded into the ground, and so on, all interact to determine whether your push makes the rock begin to roll or not. Recognition that the issue is multi-causal does *not* mean that your push is not one of the causes; in fact, it may have been a significant determiner or catalyst for the ultimate

outcome, without which the other causes would not have been activated or sufficient. Aggressive behavior, too, is multi-causal. Media violence is likely to be one of the pushes that interacts with other forces at work. In most situations, it is neither necessary nor sufficient. However, that does not mean that it is not a cause—it just means that it is *one* of the causes.

When there is a tragedy like a school shooting, we enter what could be called a “culprit mentality,” where we seek “the cause” of the tragedy (Gentile & Bushman, 2012). This is akin to the idea of “proximate cause” in law, where the goal is to assign legal responsibility for an action. The proximate cause is the last action to set off a sequence of events that produces an injury. Yet this is an inappropriate way to understand most human behavior. Human behavior is notoriously and amazingly multi-causal; there is almost never one single cause for any behavior, especially aggression. This simplistic approach to thinking about causality is misleading for both scientists and the public. Furthermore, the goal of social science is not the same as in law. Social science is concerned with all of the causes for some behavior, not only necessary, sufficient, the most recent, or the largest causes. Because media violence has been shown to increase the likelihood of aggressive behavior, it is a cause of aggressive behavior, even though it alone is not a necessary or sufficient cause.

Myth 6. Causality Means Immediacy

Many people also expect that causality requires immediacy, as in a fall causing a broken bone. As noted in the smoking and cancer example, however, physical symptoms may become visible only after some threshold of disease process is attained, which may take a long time. With regard to media violence, many people assume that the effects must be seen in the short term in order to be caused by exposure. For example, Ferguson (2002, p. 447) states, “If media violence is a necessary and direct cause of violent behavior, a significant decline in violent crime should not be occurring unless violence in the media is also declining.” We have already seen that media violence can be a cause without being a “necessary” cause. The issue of whether it is a “direct” cause seems to be the relationship between the amount of media violence and the incidence of violence in society. From the 1950s until about 1993, both the amount of media violence and the number of aggravated assaults rose in the United States (Grossman, 1996). In the latter half of the 1990s, the aggravated assault rate fell somewhat, while the amount of media violence stayed constant or increased (especially in video games; see chapters 5 and 8). This was taken by many to be “evidence” that media violence does not cause aggressive behavior.

Yet many causes have long-term effects. Consider smoking and lung cancer. Consider water, salt, and your car. Over many years, cars that are repeatedly exposed to salt rust at a higher rate than those that are not. But if you pour salt water on your car, will you see it rust in the next month or even year? No, because it is a long-term effect. Some researchers have presented evidence that the effects of media violence may be long term as well. For example, Centerwall (1989) has documented that the murder rate appears to double about 15 years after the initial introduction of television to communities /countries. It has been hypothesized that about 15 years must elapse before the full effect is revealed, as that is the amount of time it takes for a generation to grow up with the violent media and to reach prime crime-committing age. If this hypothesis is correct, then we shouldn’t expect to see immediate effects. To the extent that we expect causation to appear as immediate or short-term effects, we may miss a number of important long-term effects.

Myth 7. Effects Must Be “Big” to Be Important

Many people have agreed that the accumulated research shows that there is a systematic effect of violent media on aggressive behavior, yet also insisted that it is not a large enough effect to be important. These discussions often include a statistical approach. Meta-analyses document that the amount of variance accounted for by media violence is somewhere between 1 and 10 percent (see Comstock, Scharrer, & Powers, this volume). This means that if we drew a circle representing all the reasons why someone might act violently, media violence would account for between 1 and 10

percent of the pie. (It should be noted that some meta-analyses have reported larger numbers, and that there are a number of methodological reasons why these numbers may be underestimates; see Paik & Comstock, 1994, and Gentile & Bushman, 2012.) Several critics have argued that these effect sizes are “small and lack practical significance” (e.g., Ferguson, 2002; Freedman, 2002), but it is unclear on what basis it is made. In epidemiological terms, if only 1 percent of the people watching a violent TV show become more aggressive, and one million people watch the program, then 10,000 people were made more aggressive. That does not seem to “lack practical significance.” Indeed, many (if not most) medical studies on the effects of drugs or diet are concerned with such small effects. Supplementing one’s diet with calcium can increase bone mass, but the effect is “only” about 1 percent (Bushman & Anderson, 2001). Hormone replacement therapy in women may increase the odds of breast cancer, but across the whole population, the effect is probably less than a 1 percent effect. A daily aspirin may reduce the risk of heart attacks, but again, the effect is probably less than 1 percent. The medical profession regards these small effects as clearly important with a great deal of practical significance.

In fact, there are probably hundreds of reasons for any aggressive act (e.g., abuse, poverty, provocation, gang membership, broken home, drug use, etc.). If there are hundreds of reasons, then any single one of them should not account for much variance. That media violence consistently appears to account for at least 1 to 10 percent of the effect strikes me as surprisingly large. The U.S. Surgeon General’s report on youth violence places exposure to TV violence as just as strong a risk factor for aggressive behavior as low IQ, and more important than coming from a broken home or having abusive parents (U.S. Surgeon General, 2001). These risk factors also do not have immediate or necessarily extreme effects on aggression, but that does not render them unimportant.

Myth 8. My Child Knows the Difference between Fantasy and Reality

As noted later in the developmental tasks section, young children have a very different understanding of real and pretend than adults do. Your four-year-old may be able to say to you that “monsters aren’t real,” but she still worries that they will “get” her. This is developmentally accurate up until about age 7, after which children’s conception of fantasy and reality becomes similar to adults’. This developmental truth has led many people to believe that therefore media violence only influences people who can’t tell the difference between fantasy and reality. Unfortunately, this is not correct. I am certain that all of the children and college students who participate in media violence studies know that the violence they are seeing in the TV show or video game is not real. Nonetheless, they tend to behave more aggressively after seeing it. Similarly, adults know that advertising is fake. Nonetheless, it still works on us. Knowing that media violence is not “real” does not inoculate viewers from the effects.

Myth 9. Media Violence Helps People to Be Less Aggressive Because It Allows Them to Blow off Steam

Hundreds of studies and multiple meta-analyses (cf., [chapter 13](#), this volume; Allen, D’Alessio, and Brezgel, 1995; Anderson, 2004; Anderson & Bushman, 2001; Anderson et al., 2003; Anderson et al., 2010; Anderson, 1977; Bushman & Anderson, 2001; Bushman & Huesmann, 2006; Ferguson, 2007a; Ferguson, 2007b; Ferguson and Kilburn, 2009; Hearold, 1986; Hogben, 1998; Paik and Comstock, 1994; Savage and Yancey, 2008; Sherry, 2001; Wood, Wong, and Chachere, 1991) have consistently demonstrated that media violence exposure increases the risk of aggression. This consistency has led several prominent public health organizations (e.g., the American Medical Association, the American Academy of Pediatrics, two U.S. Surgeons General, the American Psychological Association, among others) to conclude that media violence is one causal risk factor for aggression. Despite this apparent consistency, there is still a common belief that playing a violent video game or watching a violent movie can lower the risk of future aggression by allowing the viewer to “vent” their aggression in a socially acceptable way (known as the aggression catharsis hypothesis).

Although it is an elegant idea that feels right, particularly after we have just finished playing that violent game, there are several serious problems with the catharsis hypothesis (for a detailed discussion, see Gentile, 2013). First, the concept is based on a hydraulic drive metaphor, where the pressure builds and builds and needs to “vent” or “blow off steam” in order to keep the system from exploding. Humans do have these types of drives, such as hunger and thirst, where the pressure to eat gets greater and greater and if not reduced by eating, then the system explodes (i.e., you die). Aggression is not like that. We do not become more and more aggressive and need to let it out or else we will die from not aggressing. Second, as demonstrated in most of the chapters of this book and the public health consensus above, the scientific evidence actually points in the opposite direction. Third, and most damning for the catharsis argument, it isn’t how the brain works. Every repetition of something helps us learn it better—it doesn’t remove it from the brain. Therefore, practicing aggressive thoughts and scripts by watching media violence can only *increase* how accessible and acceptable they are, not decrease them.

SUMMARY

These persistent myths underscore the importance of thinking carefully about what the effects of media violence on individuals may be. We must understand that everyone may be affected, yet not in the same way. We must recognize that most children may be affected, although we may only notice the effects in extreme cases. Two developmental theoretical approaches show a great deal of promise for helping researchers to understand the effects of media violence on children: the developmental tasks approach, and the risk and resilience approach. Each is described below, with emphasis on how they relate to media violence.

DEVELOPMENTAL TASKS APPROACH

Overview

Most children learn to talk. Most children become attached to a primary caregiver. Most children develop relationships with peers. Each of these capacities—language acquisition, development of attachment relationships, and the formation of peer relationships—is a *developmental task*—a capacity or skill that is important for concurrent and future adaptation (Sroufe, 1979). Masten and Braswell (1991, p. 13) defined developmental tasks in the following manner:

In developmental psychopathology, adaptation is often defined in terms of developmental tasks. ... The basic idea is that in order for a person to adapt, there are developmental challenges that must be met. Some arise through biological maturation, others are imposed by families and society, while others arise from the developing self.

All children of a particular society are presumed to face these tasks at some point in development; thus, these tasks serve as a barometer from which to infer competence (Masten & Coatsworth, 1998). The developmental task approach also provides researchers and practitioners with a framework for understanding how development unfolds over the course of childhood.

There is a hierarchy to these tasks (Sroufe, 1979; 1995). Different issues rise in importance depending on the developmental level of the child. Thus, for infants, the most important task that must be negotiated is developing a trusting relationship with a primary caregiver. As can be seen in [Table 2.1](#), this task recedes in importance as other tasks arise.

This does not mean, however, that early tasks are irrelevant at later stages; to the contrary, later tasks are *contingent* on the success with which earlier tasks were negotiated. That is, any measure of competence is implicitly measuring the totality of adaptation that occurred prior to that measurement. This idea—that development is cumulative and builds on prior adaptation—can also be seen in [Table 2.1](#). Here, the degree to which a child is able to form a trusting relationship with a primary caregiver has direct implications for how she negotiates the next tasks, such as active exploration of her environment. If a child has established a healthy sense of trust and this helped her to actively explore

her environment, then the child is in a good position to deal with issues of self-regulation, which are typically encountered in the preschool period. Development proceeds in this way, building on past resolutions and negotiations.

Table 2.1
Examples of Developmental Tasks

<i>Key Developmental Tasks of Infancy (0 to 12 Months)</i> ◆ Attachment to caregiver(s) ◆ Regularity of patterns ◆ Transition from reflex to voluntary behavior
<i>Key Developmental Tasks of Toddlerhood (1 to 2½ Years)</i> ◆ Curiosity, exploration, and mastery ◆ Differentiation of self from world ◆ Independence of actions, such as self-care and feeding ◆ Learning of language
<i>Key Developmental Tasks of Early Childhood (2½ to 5 Years)</i> ◆ Learning behavioral self-control and compliance with external rules ◆ Learning emotional self-control ◆ Learning gender roles and stereotypes
<i>Key Developmental Tasks of Middle Childhood (6 to 12 Years)</i> ◆ Learning how to build loyal friendships and to be accepted by peers ◆ Learning social rules and norms ◆ Adjusting to school ◆ Learning the importance of academic achievement and real-world competence ◆ Moral development ◆ Consolidating self-concept (in terms of the peer group)
<i>Key Developmental Tasks of Adolescence (13 to 18 Years)</i> ◆ Learning to build intimate and committed friendships/relationships ◆ Adjustment to pubertal changes ◆ Transition to secondary schooling ◆ Develop strong and coherent personal identity

Source: Adapted from Aber & Jones (1997), Masten & Braswell (1991), Sroufe, Cooper, & DeHart (1996), Sroufe, Egeland, & Carlson (1999).

The successful negotiation of earlier tasks sets the child on probabilistic pathways for future competence, and these can change depending on the severity of contemporaneous circumstances (e.g., parental death; Cicchetti & Toth, 1998). The implication of this is that it is erroneous to think of adaptation (or maladaptation) as something a child “has.” Instead, adaptation is a dynamic process, predicated both on past history and current context. Although change is possible, it is constrained by prior adaptation (Erikson, 1963; Sroufe, 1997). The longer a child is on an adaptive pathway (i.e., successful negotiation of prior developmental issues), the less likely it is that dire current circumstances can bump the child onto a maladaptive pathway.

Summary of Major Developmental Tasks

The effects that violent (or other) media may have on children and youth may be very different depending on the age of the child in question. As children face different developmental tasks, media are likely to have a greater or lesser effect depending on the specific issues the children are facing at that time. A brief summary of the key developmental tasks at each of five ages is presented below. These are adapted from Masten & Braswell (1991), Sroufe, Cooper, & DeHart (1996), Sroufe, Egeland, & Carlson (1999), and Aber & Jones (1997).

Infancy (approximately birth to 12 months)

During infancy, developing a trusting relationship with a caregiver is the key developmental task for healthy development. In physical development, the brain is undergoing a tremendous amount of neural network development. In cognitive development, infants exhibit learning by habituation, discrimination, classical conditioning, operant conditioning, and imitative learning. The expression of emotions begins to develop, and we see the beginnings of emotional regulation.

Toddlerhood (approximately 1 to 2½ years)

During toddlerhood, children develop a number of capacities that could be affected by media. In cognitive development, children at this age develop the capacity for symbolic representation, including language. Children also grow in their ability to use language in a competent communicative manner to conduct conversations in a socially appropriate and culturally specific manner. Social gestures begin to emerge, including conventional social gestures and symbolic gestures. Children also begin to understand themselves as distinct from others. However, children's cognition is still constrained by limited memory abilities, a lack of logic, and a difficulty distinguishing what is real and what is fantasy.

In social development, children's independence of action and feelings of competence is particularly important during toddlerhood. This is also the period where children begin to be expected to learn to regulate and control their behaviors and expressions of emotions. Toddlers begin to acquire the rules, norms, and values of society through socialization processes. Children begin to look to others for cues about how to act in new or ambiguous situations, and begin to internalize the rules and values.

In emotional development, the so-called "self-conscious" emotions such as shame, guilt, and pride emerge. Early attachment relations and the further development of those attachment relations continue to be important.

Early childhood (approximately 2½ to 5 years)

In cognitive development, children at this age begin to learn to classify things by shared characteristics, such as color, size, and shape (classification). They also begin to be able to organize things along a particular dimension, such as size or height (seriation). Learning to deploy attention with intention begins at this age, although "the tasks of selecting information to attend to, staying focused on it, and ignoring irrelevant stimuli all pose challenges to pre-schoolers" (Sroufe et al., 1996, p. 348). Children at this age continue to have difficulty solving appearance-reality problems, and reality is usually defined by the surface appearances of things. In addition, children at this age tend to only be able to focus on one piece of information at a time.

In social development, children of this age begin to develop what has been called a theory of mind. The idea is that preschoolers begin to understand that some things happen that cannot be directly observed, such as the idea that other people can make errors. However, preschoolers continue to have difficulty differentiating their own point of view from that of others.

Children at this age begin to learn "scripts" for types of behaviors, such as what happens in restaurants or what happens to get ready for bed. Similarly, preschoolers also begin acquiring a gender-role concept and to conform to sex-typed behavior. Related to this, preschoolers begin to explore adult roles in their play, including identifying with adults and mimicking adult attitudes and behaviors.

However, probably the most important developmental task for early childhood is learning self-control and self-regulation, including reflecting on one's actions, delaying gratification, tolerating frustration, and adjusting or inhibiting one's behaviors to suit the particular situational demands. These actions are part of the preschooler's growing social competence, where children begin to be able to coordinate and sustain interactions with individuals and groups of peers.

In emotional development, preschoolers begin to regulate their own emotions, including learning to be aware of the standards for behavior and using those standards to guide their words and actions. This internalization of standards is a critical part of learning to be able to feel genuine guilt or pride.

True empathy and aggression begin at this age, by which we mean actions that have no other purpose than to commiserate with another person or to cause the other person harm or distress.

In moral development, children enter Kohlberg's stage 1 of preconventional moral reasoning, in which "good" behavior is based on a desire to avoid punishments from external authorities.

Middle childhood (approximately 6 to 12 years)

In cognitive development, children at this age begin to understand the distinction between appearance and reality and to look at more than one aspect of things at the same time. They also gain a sense of industry, which Sroufe et al. (1996) defined as a basic belief in one's competence, coupled with a tendency to initiate activities, seek out learning experiences, and work hard to accomplish goals. Ideally these would lead to a sense of personal effectiveness.

In social development, learning how to form friendships is probably the main developmental task of middle childhood. This includes learning how to be part of a peer group and how to learn and adhere to the group norms. These foster the development of the self concept, in which one's sense of self is defined in part by the context of the peer group to which one belongs.

These peer relations are also important for moral development, in that the peer groups help to impart cultural norms and values. They also provide opportunities for children to see other points of view and to grow in understanding of emotion and empathy for others. Children continue to develop through Kohlberg's preconventional moral reasoning (stage 2, in which actions are motivated by desires for rewards more than desires to avoid punishment) and begin conventional moral reasoning (stage 3, in which the child's goal is to act in ways others will approve of and will avoid disapproval). Although the peer group is important as part of the engine of moral development, it is important to remember that peer groups exist within cultures and usually reflect those cultures. In fact, Sroufe et al. (1996, p. 472) have stated clearly that "the particular moral principles that children adopt are largely a product of their culture."

Adolescence (approximately 13 to 18 years)

In cognitive development, adolescents gain the ability to think about abstract concepts and relationships among abstract concepts. Attention skills also make major gains during adolescence.

In social development, the main developmental task is probably learning how to achieve deep levels of trust and closeness with both same-sex and opposite-sex peers. At home, adolescents gain more autonomy and responsibility for homework, finances, jobs, and choices affecting their futures. Personal identity makes additional gains during adolescence, in which adolescents begin to find a fit for themselves within the larger social context as well as defining themselves as unique and independent of their peer groups. Body image also begins to become important as children adjust to pubertal changes.

In moral development, adolescents continue through Kohlberg's conventional moral reasoning (stage 4, in which actions are defined as good to the extent that they perform one's duties as prescribed by the laws of society), and may begin to develop into Kohlberg's postconventional moral reasoning stages.

Media violence and developmental tasks example

Using a developmental tasks approach can help guide research and theories about media effects in a number of ways. When asking the question, "How will this show/game/movie affect children?" it becomes clear that the answer is not unidimensional. The effects are likely to differ greatly depending on the age of the child. Consider the following example taken from a nationally broadcast episode of professional wrestling (WWF *Smackdown*, Oct 7, 1999). Wrestling was selected as an example of media violence here because it is highly watched by children (it has historically been the highest rated show on cable; Keller, 2002).

A male wrestler, Jeff Jarrett, is angry at his wrestler girlfriend, Miss Kitty, because she lost a wrestling match the previous week. In order to “get back on [his] good side,” he requires Miss Kitty to participate in a mud wrestling match. He asks the reigning ladies’ champion, Ivory, to stand near the ring to watch. He then announces that the goal of the match is to remove the opponent’s shirt and bra in order to win. He then throws Ivory into the mud, to her apparent surprise. Miss Kitty immediately attacks Ivory, removing Ivory’s dress. Meanwhile, Jarrett makes comments about women being the lowest form of life and the announcers make lascivious comments about the women’s bodies. Ivory eventually removes Miss Kitty’s bikini top, “winning” the match. Incensed older women wrestlers arrive to confront Jarrett, who promptly throws them into the mud, while making comments about them being fat old sows. Ultimately, another lady wrestler, Chyna, sneaks up behind Jarrett, and pushes him into the mud. How would children who watch shows like this be affected? The developmental tasks approach provides a framework to understand how children may be affected at different ages.

For infants, it is unlikely to have much effect, unless the parents watch programs like this so much that it interferes with their ability to care for the infant or disrupts the infant’s ability to set regular patterns. For toddlers, who are beginning to use language, there were a number of derogatory terms used that children might learn. However, children at this age are just beginning to acquire the standards and values of society through the socialization process. This type of program shows violence as the solution to interpersonal conflict, as well as the “normality” of verbal and physical abuse toward women (especially scantily clad or nude women). Habituation and desensitization processes have begun.

In early childhood, where the main developmental tasks are about behavioral self-control, emotional self-control, and gender roles, this type of program may have a number of negative effects. Very little self-control is displayed. Words are not used to resolve problems but only to enhance the problems. In this specific example, the male was the one with all the power: he set the rules for the engagement, and even though Ivory was “tricked” into the ring, there was no reason she needed to comply with Jarrett’s rules—yet she did. The entire episode was derogatory toward women, and even though some might say that the women won in the end, they did it on his terms, not theirs. Children at this age may begin to see women as needing to do whatever men say they should to gain the approval of men and that such behaviors are normal or natural.

In middle childhood, social rules and norms take on increased importance, and they are likely to learn lessons about the importance of physical domination and humiliation of others as an acceptable method of conflict resolution—that is, if they have not already habituated to this level of physical and verbal abuse and therefore do not see it at all. Furthermore, in this example, competence was defined only in terms of ability to fight (although there is also a subtext of sexuality as competence).

In adolescence, the major developmental task is learning how to have intimate and committed relationships. This type of show portrays the relationships between men and women very stereotypically, where the male has the power and the females are submissive. Furthermore, it portrays physical aggression between the sexes as acceptable (and sexual). By this developmental stage, physical and verbal violence in the media will likely appear unremarkable—a natural part of our culture and only a mirror of our society.

We do not mean to suggest that watching one episode of any program is likely to have a large, immediate effect. But any immediate effects as well long-term effects are likely to be different based on the age of the child, and the developmental tasks approach provides a framework for designing and testing hypotheses about the types of effects we might expect at different ages. This approach demonstrates why expecting media violence to affect everyone the same way (myth 4 above) is inappropriate—media violence *shouldn’t* affect children of different ages the same way.

A RISK AND RESILIENCE APPROACH TO DEVELOPMENT

The developmental tasks approach focuses on normative development; that is, every child will go through these phases with varying degrees of ease. In contrast, the risk and resilience approach (also

called a risk and protective factor approach) focuses on the unique pattern of risk and protective factors for each individual child (see reviews by Glantz & Johnson, 1999; Masten & Coatsworth, 1998; Prot & Gentile, 2014). Risk factors are those that put a child at risk for maladaptive development and poor outcomes, whereas protective factors are those that help to limit the risk or maintain resilience to risk factors. These risk and protective factors can exist at many different levels, including genetic, personality traits, prior trauma, attitudes, family variables, peer support, neighborhood context, cultural context, economics, etc. With regard to aggression, the U.S. Surgeon General's report on youth violence (2001) defines risk factors as "personal characteristics or environmental conditions that predict the onset, continuity, or escalation of violence" (p. 58).

This approach helps to explain why we see greater effects of media violence on some children than on others. Exposure to media violence is likely to be a risk factor for all children, increasing the risk for aggressive behaviors. However, some children have additional risk factors that enhance the effects of media violence exposure, while other children may have protective factors that attenuate the risk of aggressive behavior. This could make it look like only some children are affected by media violence, when in fact they all are affected; it's just much easier to notice the effect in some children.

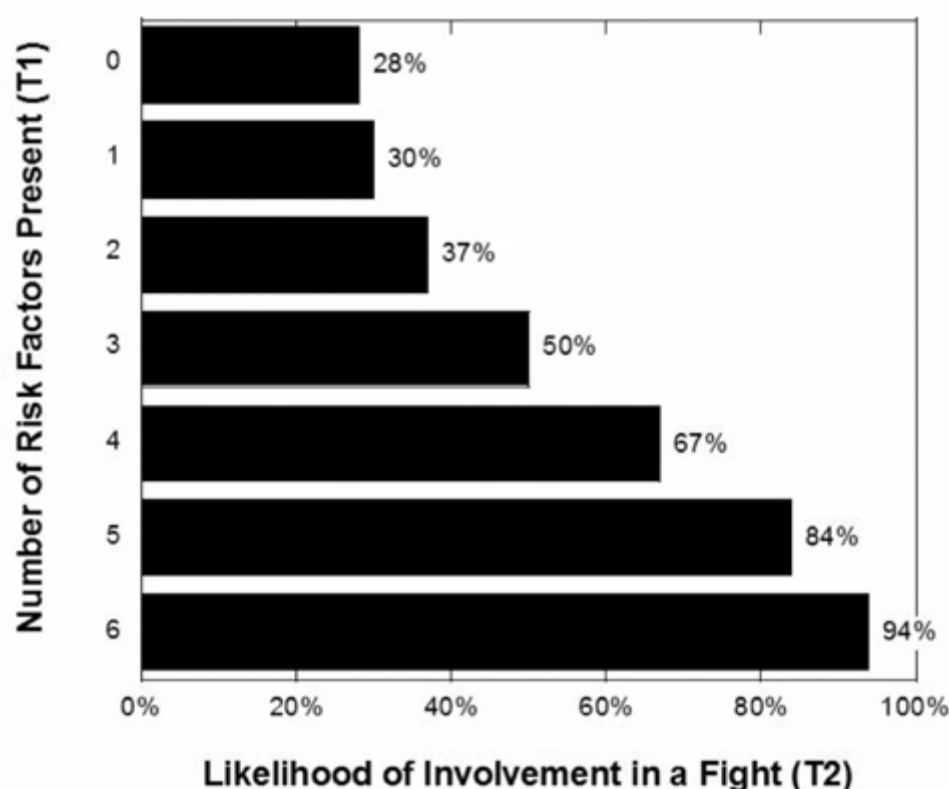
In the decade since the first edition of this book, much progress has been made in risk and protective factor models, and researchers are beginning to use them to predict aggression. Four primary types of models have been identified: cascade effects, dose-response gradients, pathway models, and turning-point models. (Due to space limitations, these are described only briefly; see Prot & Gentile [2014] for more detail and examples of studies.)

Cascade effects are where small changes in one area of life can "snowball" to spread across areas, across time, across people, and even across generations (Masten & Narayan, 2012). Disruptions in adaptive behavior can spread and have negative consequences in multiple domains, whereas competence can act as scaffolding on which competence in other domains develops (Masten, Burt, & Coatsworth, 2006). For example, Slater, Henry, Swaim, & Anderson (2003) measured students in 20 middle schools four times over two years. They found that aggressive children seek out media violence, but that media violence exposure predicted becoming more aggressive, which further made children seek out more media violence, etc., in a "downward spiral," each influencing the other across development.

Dose-response gradients demonstrate that as the level of risk rises, the likelihood for problems also rises. This may not always happen in a linear fashion, but can also include thresholds, asymptotic patterns, or inverted-U models (Luthar, Cicchetti, & Becker, 2000; Masten, 2012). For example, Gentile and Bushman (2012) examined the cumulative effects of six different risk and protective factors on children's aggressive behavior over a six-month period. Predictors included media violence exposure, physical victimization, participant sex, hostile attribution bias, parental monitoring, and prior aggressive behavior. Each risk factor at Time 1 predicted an increased risk of physical aggression at Time 2, whereas each protective factor predicted a decreased risk of physical aggression at Time 2. The combination of risk factors, however, was a better predictor than any individual risk factor (Figure 2.3). Importantly, they found both linear and multiplicative effects of multiple risk factors.

Figure 2.3

Predicted Likelihood of Time 2 Fights from Number of Risk Factors Present at Time 1, Demonstrating Nonlinear Cumulative Risk



Source: Gentile & Bushman, 2012.

Pathway models describe potential patterns of functioning before, during, and after exposure to some adverse event (Masten, 2011). For example, Krahé, Busching, and Möller (2012) studied 1,715 middle school students and found that across a two-year period there were three types of students: those who consistently did not consume much media violence, those who consistently watched a lot, and those who started high but decreased their media violence exposure. These three trajectories also predicted the trajectory of physical aggression. Stable low media violence exposure predicted stable low aggression. Stable high media violence exposure predicted a small increase in aggression. Decreasing media violence exposure predicted a decrease in aggressive behavior.

Turning point models attempt to explain why some people may seemingly suddenly shift from a bad trajectory to a positive one. Late adolescence/emerging adulthood has been found to be a transitional time during which such cases are more likely, because this period of development provides new opportunities, motivations, and impulse control (Burt & Masten, 2010; Burt & Paysnick, 2012; Masten & Tellegen, 2012). Furthermore, adolescents are more willing to take risks due to changes the adolescent brain is undergoing (Spear, 2000). No media violence studies have yet focused specifically on this approach, although the group described above who stopped viewing media violence and consequently reduced their aggression may be such an example (Krahé et al., 2012).

With regard to media violence, exposure to entertainment media violence acts like a risk factor for aggressive behavior and other negative outcomes. The presence of this single risk factor is not sufficient to cause children to hit someone, and certainly not sufficient to cause someone to pick up a gun and begin shooting. However, each additional risk factor children have for aggressive behavior (e.g., being male, having been bullied, low parental monitoring, poverty, gang membership, drug use, history of being abused, access to guns, etc.) increases the risk of that child acting aggressively. In contrast, each additional protective factor children have (e.g., stable family environment, good school performance, open communication with parents, etc.) decreases the risk of aggressive behavior.

The risk and resilience approach helps to answer the comment most people make about media violence: “I watched a lot of media violence as a kid, and I never shot anyone.” We need to remember that “shooting someone” is a highly extreme behavior. Most people will never engage in such an extreme behavior, and exposure to media violence is not such a powerful effect as to be able to make such extreme changes in people. One possible metaphor for this process is to consider the

thermometer shown in [Figure 2.4](#).¹ At the lowest end, a child's behavior is routinely respectful and polite. At the highest end, a child engages in the ultimate disrespectful behavior of shooting someone. It is likely that regular exposure to media violence might be able to shift someone about two or three spots on the thermometer. It certainly isn't a strong enough effect to shift someone from routinely respectful behavior to shooting someone, but it might change someone such that he or she begins showing rude and verbally aggressive behavior. If, however, an individual starts out with other risk factors for violent behavior, and is already at the verbally aggressive spot on the thermometer, regular exposure to media violence may just add enough additional risk to get him to start pushing and shoving others around. The child may also have additional protective factors that help to keep the level lower.

We have described a number of commonly held beliefs about media effects, and how those beliefs can hamper the ability to accurately predict and interpret the effects of violent media portrayals on children. These myths include the beliefs that media effects must be simple, direct, obvious, severe, and affect everyone the same way. They also include errors in the idea of what a "cause" must be, or how large an effect must be to be important. Although the research is clear that media violence can have a negative impact on children, it has been less clear why some children may show a larger effect than others, or why some children may be affected in different ways. The persistence of the myths has made it difficult to understand why we should not expect children to be affected identically. The developmental tasks approach helps to describe why children at different developmental "stages" would be expected to be influenced differently. The risk and resilience approach helps to describe why children at any given age could be affected the same way, but one child would show the effects behaviorally and another might not. These two developmental approaches have great promise for the field of media effects research, as they help us to understand why children may be affected differently by exposure to media violence, and also why even though most children will not become seriously violent from exposure to media violence, they may nonetheless be affected in an important manner.

Figure 2.4
The Metaphorical Aggression Thermometer



Source: Gentile & Bushman, 2012.

NOTE

1. This metaphor is used in the understanding that it is imperfect. For example, it is not clear that relational aggression is less aggressive than physical aggression, as the consequences are sometimes worse (see Coyne & Stockdale, this volume). It is intended solely to help describe how different risk factors can increase the likelihood of more severe aggressive behaviors, and how protective factors can decrease the likelihood of aggressive behaviors.

REFERENCES

- Aber, J. L., & Jones, S. J. (1997). Indicators of positive development in early childhood: Improving concepts and measures. In R. M. Hauser, B. V. Brown, & W. R. Prosser (Eds.), *Indicators of children's well-being* (pp. 395–408). New York, NY: Sage Foundation.
- Allen, M., D'Alessio, D., & Brezgel, K. (1995). A meta-analysis summarizing the effects of pornography II: Aggression after exposure. *Human Communication Research*, 22, 258–283.
- Anderson, C. A. (2004). An update on the effects of playing violent video games. *Journal of Adolescence*, 27, 113–122.
- Anderson, C. A., Berkowitz, L., Donnerstein, E., Huesmann, L. R., Johnson, J. D., Linz, D., et al., (2003). The influence of media violence on youth. *Psychological Science in the Public Interest*, 4, 81–110.
- Anderson, C. A., Shibuya, A., Ihori, N., Swing, E. L., Bushman, B. J., Sakamoto, A., et al., (2010). Violent video game effects on aggression, empathy, and prosocial behavior in Eastern and Western countries: A meta-analytic review. *Psychological Bulletin*, 136(2), 151–173. doi:10.1037/a0018251
- Andison, F. S. (1977). TV violence and viewer aggression: A cumulation of study results. *Public Opinion Quarterly*, 41, 314–331.
- Brachear, M. (2002, July 19). Copycat stunt injures youth. *Dallas Morning News*. <http://www.dallasnews.com/latestnews/stories/071902dntextstunt.811a0.html>
- Burt, K. B., & Masten, A. S. (2010). Development in the transition to adulthood: Vulnerabilities and opportunities. In J. E. Grant & M. N. Potenza (Eds.), *Young adult mental health* (pp. 5–18). New York, NY: Oxford University Press.
- Burt, K. B., & Paysnick, A. A. (2012). Resilience in the transition to adulthood. *Development and Psychopathology*, 24(2), 493–505.
- Bushman, B. J., & Anderson, C. A. (2001). Media violence and the American public. *American Psychologist*, 56, 477–489.
- Bushman, B. J., & Huesmann, L. R. (2006). Short-term and long-term effects of violent media on aggression in children and adults. *Arch Pediatr Adolesc Med*, 160(4), 348–352.
- Centerwall, B. S. (1989). Exposure to television as a risk factor for violence. *American Journal of Epidemiology*, 129, 643–652.
- Cicchetti, D., & Toth, S. L. (1998). The development of depression in children and adolescents. *American Psychologist*, 53, 221–241.
- Complete coverage on theater shooting in Aurora. www.cnn.com/SPECIALS/us/colorado-shooting
- Dandekar, P., Goel, A., & Lee, D. T. (2013). Biased assimilation, homophily, and the dynamics of polarization. *Proceedings of the National Academy of Sciences*, 110(15), 5791–5796. doi:www.pnas.org/cgi/doi/10.1073/pnas.1217220110
- Donnerstein, E., Slaby, R. G., & Eron, L. D. (1994). The mass media and youth aggression. In L. D. Eron, J. H. Gentry, & P. Schlegel (Eds.), *Reason to hope: A psychosocial perspective on violence and youth* (pp. 219–250). Washington, DC: American Psychological Association.
- Erikson, E. H. (1963). *Childhood and society* (2nd ed.). New York, NY: Norton.
- Ferguson, C. J. (2002). Media violence: Miscast causality. *American Psychologist*, 57, 446–447.
- Ferguson, C. J. (2007a). Evidence for publication bias in video game violence effects literature: A meta-analytic review. *Aggression and Violent Behavior*, 12(4), 470–482.
- Ferguson, C. J. (2007b). The good, the bad and the ugly: A meta-analytic review of positive and negative effects of violent video games. *Psychiatric Quarterly*, 78, 309–316.
- Ferguson, C. J. (2010). Blazing angels or resident evil? Can violent video games be a force for good? *Review of General Psychology*, 14, 68–81.

- Ferguson, C. J., & Kilburn, J. (2009). The public health risks of media violence: A meta-analytic review. *The Journal of Pediatrics*, 154, 759–763.
- Freedman, J. L. (2002). *Media violence and its effect on aggression*. Toronto, ON: University of Toronto Press.
- Gentile, D. A. (2013). Catharsis and media violence: A conceptual analysis. *Societies*, 3, 491–510. doi:10.3390/soc3040491
- Gentile, D. A., & Bushman, B. J. (2012). Reassessing media violence effects using a risk and resilience approach to understanding aggression. *Psychology of Popular Media Culture*, 1(3), 138–151.
- Gentile, D. A., & Sesma, A. (2003). Developmental approaches to understanding media effects on individuals. In D. A. Gentile (Ed.), *Media violence and children: A complete guide for parents and professionals*. Westport, CT: Praeger.
- Gibbs, N., & Roche, T. (1999, December 20). The Columbine tapes. *Time*, 40–50.
- Glantz, M. D., & Johnson, J. L. (Eds.). (1999). *Resilience and development: Positive life adaptations*. New York, NY: Kluwer.
- Greene, V. S. (1999). *Television viewing, perceptions of advertising's influence and appearance-related concerns*. Unpublished manuscript, University of Massachusetts, Amherst.
- Grossman, D. (1996). *On killing*. Boston, MA: Little, Brown & Co.
- Hearold, S. (1986). A synthesis of 1043 effects of television on social behavior. In G. Comstock (Ed.), *Public communication and behavior* (Vol. 1, pp. 65–133). New York: Academic Press.
- Hogben, M. (1998). Factors moderating the effect of television aggression on viewer behavior. *Communication Research*, 25, 220–247.
- Keller, W. (2002, August 10). *Wrestling's state of business is taking more knocks than deserved*. http://www.pwtorch.com/artman/publish/printer_163.shtml
- Krahé, B., Busching, R., & Möller, I. (2012). Media violence use and aggression among German adolescents: Associations and trajectories of change in a three-wave longitudinal study. *Psychology of Popular Media Culture*, 1, 152–166.
- Luthar, S. S., Cicchetti, D., & Becker, B. (2000). The construct of resilience: A critical evaluation and guidelines for future work. *Child Development*, 71, 543–562.
- Masten, A. S. (2011). Resilience in children threatened by extreme adversity: Frameworks for research, practice, and translational synergy. *Development and Psychopathology*, 23, 141–154.
- Masten, A. S. (2012). Risk and resilience in the educational success of homeless and highly mobile children: Introduction to the special section. *Educational Researcher*, 41, 363–365.
- Masten, A. S., & Braswell, L. (1991). Developmental psychopathology: An integrative framework. In P. R. Martin (Ed.), *Handbook of behavior therapy and psychological science: An integrative approach* (pp. 35–56). New York, NY: Pergamon Press.
- Masten, A. S., Burt, K. B., & Coatsworth, J. D. (2006). Competence and psychopathology in development. In D. Cicchetti & D. J. Cohen (Eds.), *Developmental psychopathology* (Vol. 3, 2nd ed., pp. 696–738). Hoboken, NJ: Wiley.
- Masten, A. S., & Coatsworth, J. D. (1998). The development of competence in favorable and unfavorable environments: Lessons from research on successful children. *American Psychologist*, 53, 205–220.
- Masten, A. S., & Narayan, A. J. (2012). Child development in the context of disaster, war and terrorism: Pathways of risk and resilience. *Annual Review of Psychology*, 63, 227–257.
- Masten, A. S., & Tellegen, A. (2012). Resilience in developmental psychopathology: Contributions of the Project Competence Longitudinal Study. *Development and Psychopathology*, 24, 345–361.
- Paik, H., & Comstock, G. (1994). The effects of television violence on antisocial behavior: A meta-analysis. *Communication Research*, 21(4), 516–546.
- Potter, W. J. (1999). *On media violence*. Thousand Oaks, CA: Sage Publications.
- Prot, S., & Gentile, D. A. (2014). Applying risk and resilience models to predicting the effects of media violence on development. *Advances in Child Development and Behavior*, 46, 215–244.
- Savage, J., & Yancey, C. (2008). The effects of media violence exposure on criminal aggression: A meta-analysis. *Criminal Justice and Behavior*, 35, 772–791.
- Sherry, J. L. (2001). The effects of violent video games on aggression. *Human Communication Research*, 27, 409–431.
- Slater, M. D., Henry, K. L., Swaim, R. C., & Anderson, L. L. (2003). Violent media content and aggressiveness in adolescents: A downward spiral model. *Communication Research*, 30, 713–736.

- Spear, L. P. (2000). The adolescent brain and age-related behavioral manifestations. *Neuroscience and Biobehavioral Reviews*, 24, 417–463.
- Sroufe, L. A. (1979). The coherence of individual development: Early care, attachment, and subsequent developmental issues. *American Psychologist*, 34, 834–841.
- Sroufe, L. A. (1995). *Emotional development: The organization of emotional life in the early years*. New York, NY: Cambridge Press.
- Sroufe, L. A. (1997). Psychopathology as an outcome of development. *Development and Psychopathology*, 9, 251–268.
- Sroufe, L. A., Cooper, R. G., & DeHart, G. B. (1996). *Child development: Its nature and course* (3rd ed.). New York, NY: McGraw-Hill, Inc.
- Sroufe, L. A., Egeland, B., & Carlson, E. A. (1999). One social world: The integrated development of parent-child and peer relationships. In W. A. Collins & B. Laursen (Eds.), *Relationships as developmental context: The 30th Minnesota symposium on child psychology*. Hillsdale, NJ: Erlbaum.
- Stiff sentence: 14-year-old gets life in prison for wrestling death. (2001, March 9). ABC News. <http://abcnews.go.com/sections/us/DailyNews/tate-sentence010309.html>
- Texas talkback. (N.d.). <http://www.txcn.com/cgi-bin/survey.cgi?survey1104&stepView%20comments>
- U.S. Surgeon General. (2001, April 11). *Youth violence: A report of the surgeon general*. Washington, DC: United States Surgeon General.
- Walsh, D. A. (2001). *Dr. Dave's cyberhood*. New York, NY: Fireside.
- White, J. (2013, December 16). I have Asperger's, I play violent video games—and I'm not going to shoot you. *CCN.com*. <http://www.cnn.com/2013/12/13/opinion/aspergers-video-games-sandy-hook-irpt/index.html?irefallsearch>
- Wood, W., Wong, F., & Chachere, J. (1991). Effects of media violence on viewers' aggression in unconstrained social interaction. *Psychological Bulletin*, 109, 371–383.
- Woodward, G. C., & Denton, R. E. (2000). *Persuasion & influence in American life* (4th ed.). Prospect Heights, IL: Waveland Press.